3. Input parameters implied by the term-by-term solution vs. Table 1 of universe_bestfit_thinshell.pdf. Each column keeps that fit's shell assumption. Newtonian Ω_s free: $\Omega_s = 1.012$, thin shell $R_{\rm in} = D - R$, $R_{\rm out} = D + R$. Newtonian Ω_s=0.049: $R_{\rm in} = D - 1.028\,R$, $R_{\rm out} = D + 1.475\,R$. Relativistic: $R_{\rm in} = D - 1.040\,R$, $R_{\rm out} = D + 1.171\,R$. The unknowns are solved from the Sec. 2 equations (2ℓ phase $\to \ell_0$, 2ℓ amplitude $\to \beta \to M_c/M_{\rm crit}$, $1\ell \to$ Amp, φ, 3ℓ cos $\to \varepsilon Q \to D$; C_m, S_m, m = 1–3, by least squares). The $a_m(\beta,\gamma)$ come from the code's exact $h(q)$, not the linear formula. Masses use $H_0 = 70$, $\rho_s = \Omega_s \rho_c$.

	Newtonian, Ω_s free		Newtonian, Ω_s=0.049		Relativistic (LTB)
Quantity	term-by-term	Table 1	term-by-term	Table 1	Table 1
D/R	6.474	6.554	2.589	3.439	2.303
D	2.857×10^{24} km (92.6 Gpc)	2.892×10^{24} km (93.7 Gpc)	1.143×10^{24} km (37.0 Gpc)	1.517×10^{24} km (49.2 Gpc)	1.016×10^{24} km (32.9 Gpc)
$R_{\rm in}$	78.3 Gpc	79.4 Gpc	22.3 Gpc	34.5 Gpc	18.1 Gpc
$R_{\rm out}$	106.9 Gpc	108.0 Gpc	58.1 Gpc	70.3 Gpc	49.7 Gpc
M_c	2.38×10^{54} kg 1.20×10^{24} $M_\odot$	2.47×10^{54} kg 1.24×10^{24} $M_\odot$	9.41×10^{52} kg 4.73×10^{22} $M_\odot$	2.31×10^{53} kg 1.16×10^{23} $M_\odot$	3.38×10^{54} kg 1.70×10^{24} $M_\odot$
$M_c/M_{\rm crit}(D)$	0.00265	0.00265	0.00164	0.00171	0.0835
$M(<D)$, total	3.62×10^{56} kg	3.72×10^{56} kg	2.29×10^{54} kg	4.56×10^{54} kg	5.04×10^{54} kg
$M_{\rm shell}$, total	8.50×10^{56} kg	8.71×10^{56} kg	1.03×10^{55} kg	1.70×10^{55} kg	6.48×10^{54} kg
M within R	3.35×10^{54} kg	3.35×10^{54} kg	1.62×10^{53} kg	1.62×10^{53} kg	1.62×10^{53} kg
Ω_s	1.012[a]	1.012	0.049[a]	0.049	0.049
ℓ_0	13.2°	12.8°	12.8°	13.0°	13.5°
Amp	−0.1246	−0.1240	−0.0637	−0.0622	−0.0558
φ	1.267 rad	1.264 rad	0.941 rad	0.929 rad	0.848 rad
R^2 vs Migkas	0.909	0.909	0.881	0.893	0.885
R^2 vs A_F	0.987	0.987	0.970	0.977	0.963

[a] Held fixed (that column's shell assumption). The harmonic solve weights C_m, S_m equally, not the 71 points, so its R^2 can sit slightly below the point fit.

4. Model vs. generally accepted values. Model columns use the Table 1 fits. $\langle H\rangle$ is the model's sky-mean expansion rate (mean of $a_f = |\Delta\vec{V}|$/separation over the code's directions at R, with the fitted sine). The fits fix only its shape, so its absolute value is solved from Migkas et al.'s own H_0 values by fitting $H(\ell) = \langle H\rangle\, A(\ell)/\langle A\rangle$ to them (anchor: their all-combined joint result[a]). All rate-dependent rows follow from $\langle H\rangle$: rates, k, $V_O \propto \langle H\rangle$; masses, densities $\propto \langle H\rangle^2$; $t_0 \propto 1/\langle H\rangle$. Lengths use $R = 14.3$ Gpc.

Quantity	Accepted value [source]	Newt. Ω_s free (0.909)	Newt. Ω_s=0.049 (0.893)	Relativistic LTB (0.885)
H_0 [km s^{-1} Mpc^{-1}]	67.36 ± 0.54 (CMB) [1]; 73.04 ± 1.04 (local) [2]	$\langle H\rangle = 71.8\pm0.5$[a] (= $0.971\,k$)	$\langle H\rangle = 71.6\pm0.5$[a] (= $1.012\,k$)	$\langle H\rangle = 71.6\pm0.5$[a,d] (= $0.956/t_0$)
k (LTB: $1/t_0$) [km s^{-1} Mpc^{-1}]	–	74.0	70.8	74.9
Age / k^{-1}	13.797 ± 0.023 Gyr [1]	$k^{-1} = t_0 = 13.22$ Gyr	$k^{-1} = t_0 = 13.81$ Gyr	$t_0 = 13.05$ Gyr
Ordinary-matter mass within R	1.51×10^{53} kg [1][b]	3.67×10^{54} kg	1.64×10^{53} kg	1.60×10^{53} kg
Total matter mass within R	9.67×10^{53} kg [1][b]	same (no dark matter)	same (no dark matter)	same (no dark matter)
Ordinary-matter density ρ_b	4.20×10^{-28} kg m^{-3} [1]	1.02×10^{-26} kg m^{-3}	4.56×10^{-28} kg m^{-3}	4.46×10^{-28} kg m^{-3}
Ω_b (= Ω_m in the model)	$\Omega_b = 0.0493$, $\Omega_m = 0.3153 \pm 0.0073$ [1]	1.012 (1.053 vs $\langle H\rangle$)	0.049 (0.047 vs $\langle H\rangle$)	0.049 (0.046 vs $\langle H\rangle$)
Critical density ρ_c at $\langle H\rangle$	8.52×10^{-27} kg m^{-3} (H_0=67.36) [1]	9.68×10^{-27} kg m^{-3}	9.63×10^{-27} kg m^{-3}	9.64×10^{-27} kg m^{-3}
Earth (Sun) speed vs. CMB	369.82 ± 0.11 km s^{-1} [3]	$V_O = 6.23\times10^6$ km s^{-1} = $20.8\,c$	$V_O = 3.26\times10^6$ km s^{-1} = $10.9\,c$	$V_O = 2.29\times10^6$ km s^{-1} = $7.6\,c$
ratio V_O/(CMB dipole speed)	1	16,848	8,804	6,199
Local Group speed vs. CMB	620 ± 15 km s^{-1} [3]	– (not modelled)	– (not modelled)	– (not modelled)
Bulk flow within $150\,h^{-1}$ Mpc	395 ± 29 km s^{-1} [4]	6.0 km s^{-1}[c]	0.33 km s^{-1}[c]	not computed
Core mass M_c (vs. accepted M_b in R)	–	2.71×10^{54} kg (18×)	2.33×10^{53} kg (1.5×)	3.34×10^{54} kg (22×)
Mass inside D (same comparison)	–	4.07×10^{56} kg (2,692×)	4.61×10^{54} kg (30×)	4.98×10^{54} kg (33×)

[a] Weighted least-squares fit of $\langle H\rangle A(\ell)/\langle A\rangle$ to Migkas et al. (2021, A&A, 649, A151) all-combined H_0: 66.5 ± 1.0 toward $\ell = 273°$ and 72.8 ± 0.6 for the rest of the sky (model value at the cone centre, and the model mean outside a 75° cone). Across all 11 Migkas 2020/2021 choices (each relation, sample and joint result) $\langle H\rangle$ = 70.4±2.4 / 70.3±2.1 / 70.4±2.0 (mean±s.d., the three fits). Migkas's absolute H_0 values rest on their fiducial calibration ($H_0 = 70$ assumed to calibrate the scaling relations), so this anchor inherits it. [b] Derived from [1]: $\Omega_b h^2 = 0.02237$, $\Omega_m = 0.3153$, $H_0 = 67.36$, times the volume of the $R = 14.3$ Gpc sphere; ratios in brackets are model / this accepted ordinary-matter mass. [c] Mean of $\Delta\vec{V}$ over a $150\,h^{-1}$ Mpc ball around Earth, from the code's exact expansion-rate profile (second order in radius/D); the sine factor is not a velocity. [d] LTB: mean of the local expansion rate over the in-plane directions, $(H_\perp + H_r)/2$; the fit's $z = 0.3$ curve is normalised and carries no absolute rate. LTB values rescaled from the fit (which set the 3D-mean local rate to 70) with the same power laws (approximate in GR; the fitted valley shows the shape does not depend on HD/c).

Sources. [1] Planck Collaboration 2020, A&A, 641, A6 (Planck 2018 VI, TT,TE,EE+lowE+lensing). [2] Riess, A. G., et al. 2022, ApJL, 934, L7 (SH0ES). [3] Planck Collaboration 2020, A&A, 641, A1 (Planck 2018 I; Local Group value after Kogut et al. 1993, ApJ, 419, 1). [4] Watkins, R., et al. 2023, MNRAS, 524, 1885 (CosmicFlows-4). Sedov–Taylor: Taylor, G. I. 1950, Proc. R. Soc. Lond. A, 201, 159; Sedov, L. I. 1959, Similarity and Dimensional Methods in Mechanics (Academic Press).